

База:

Расстояние и-го
скрещ. прямых

1) Уравнение плоскости в $\mathbb{R}^3$

$\chi: Ax + By + Cz + D = 0$, где:

$\vec{n}(A, B, C)$ - нормаль

$A_0(x, y, z)$ - точка $\in \chi$

$$d(l_1, l_2)$$

2) Уравнение прямой

$$\frac{x-x_0}{A} = \frac{y-y_0}{B} = \frac{z-z_0}{C}, \text{ где}$$

$\vec{n}(A, B, C)$ - нормаль

параметрически:

$$\begin{cases} x = x_0 + At \\ y = y_0 + Bt \\ z = z_0 + Ct \end{cases}$$

Формулы:

угол между прямыми:

$$\cos \varphi = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{\|\vec{n}_1\| \|\vec{n}_2\|}$$

Расстояние от точки P

до прямой χ ($\vec{n}$ - ее нормаль):
 $M_0 \in \chi$

$$d = \frac{|\vec{n} \times \vec{M_0 P}|}{\|\vec{n}\|}$$

Решение:

(k)

~913

$M(1, 1, -1)$

$\vec{n}(1, -2, 3)$

$$x - 2y + 3z + D = 0$$

$$1 - 2 - 3 + D = 0$$

$$D = 4$$

$$\text{Ответ: } x - 2y + 3z + 4 = 0$$

~917

$M_1(3, 4, -5)$

a_1, a_2

$a_1 = \{3, 1, -1\}$

$a_2 = \{1, -2, 1\}$

$$\begin{vmatrix} x-3 & y-4 & z+5 \\ 3 & 1 & -1 \\ 1 & -2 & 1 \end{vmatrix} =$$

$$= (x-3)(-1) - (y-4)(1) + (z+5)(-7) =$$

$$= -x + 3 - y + 4 - 7z - 35 =$$

$$= -x + 4y - 7z - 16$$

~924

$$1) 2x - 3y + 5z - 7 = 0$$

$$2x - 3y + 5z + 3 = 0$$

непараллельны ✓

~925

$$1) 3x - y - 2z - 5 = 0 \quad \vec{n}_1(3, -1, -2)$$

$$x + y - 3z + 2 = 0 \quad \vec{n}_2(1, 1, -3)$$

$$\cos \varphi = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{\|\vec{n}_1\| \|\vec{n}_2\|} =$$

$$= \frac{|3 - 9 + 6|}{\sqrt{9+1+4} \sqrt{1+1+9}} = 0$$

$$\varphi = \arccos(0) = \frac{\pi}{2}$$

или $\perp$

~928

$$1) x - y\sqrt{2} + z - 1 = 0$$

$$x + y\sqrt{2} - z + 3 = 0$$

$$\vec{n}_1 = (1, -\sqrt{2}, 1)$$

$$\vec{n}_2 = (1, \sqrt{2}, -1)$$

$$\cos \varphi = \frac{|1 - 2 - 1|}{\sqrt{1+2+1} \sqrt{1+2+1}} = \frac{2}{4} = \frac{1}{2}$$

$$\varphi = \arccos\left(\frac{1}{2}\right)$$

$$\varphi = \frac{\pi}{3}, \frac{2\pi}{3}$$

↙

~848 найти расстояние

от точки $P(7, 9, 7)$ до

плоскости $\frac{x-2}{4} = \frac{y-1}{3} = \frac{z}{2}$

$$\vec{n} = (4, 3, 2) \quad M_0(2, 1, 0)$$

$$d = \frac{|\vec{n} \times \overrightarrow{M_0 P}|}{\|\vec{n}\|}$$

$$\overrightarrow{M_0 P} = (5, 8, 7)$$

$$= (5, -18, 17)$$

$$d = \frac{\sqrt{25 + 324 + 289}}{\sqrt{16 + 9 + 4}} = \frac{\sqrt{638}}{\sqrt{29}}$$

$$= \sqrt{22}$$

(2g)

~828

$$\frac{x-12}{4} = \frac{y-9}{3} = \frac{z-1}{1} \quad a_0(12; 9; 1)$$

$$\begin{cases} x = 12 + 4t \\ y = 9 + 3t \\ z = 1 + t \end{cases}$$

$$3x + 5y - z - 2 = 0$$

$$36 + 12t + 45 + 15t - 1 - t - 2 = 0$$

$$26t = -78$$

$$t = -3$$

$$(10; 0; -2)$$

~833

$$M_0(3, -2, 4)$$

$$\pi: 5x + 3y - 7z + 1 = 0$$

$$\vec{n} = (5, 3, -7)$$

$$Ax + By + Cz + D = 0$$

$$5x + 3y - 7z + D = 0$$

$$15 - 6 - 28 + D = 0$$

$$D = 19$$

$$5x + 3y - 7z + 19 = 0$$

~834

$$M_0(0, 0, 0)$$

$$\frac{x+2}{4} = \frac{y-3}{5} = \frac{z-1}{-2}$$

$$\vec{n}(4; 5; -2)$$

$$Ax + By + Cz + D = 0$$

$$D = 0$$

$$4x + 5y - 2z = 0$$

+ поворот координат

+ сдвигаем ось z
вправо (по там easy)

$$\vec{n}_1(-4, -2, 4)$$

$$\vec{n}_2(2, 1, -2)$$

$$\cos \varphi = \frac{|-8 - 2 - 8|}{\sqrt{32+4} \sqrt{9}} =$$

$$= \frac{|-18|}{6 \cdot 3} = 1$$

$$\varphi = \pi$$

$$F_0 = a_{10}x + a_{20}y + a_{00}$$

$$\sim 835$$

$$A(4; -3; 1)$$

$$\vec{n}(1; 2; -1)$$

$$\frac{x-4}{1} = \frac{y+3}{2} = \frac{z-1}{-1}$$

$$\vec{n} = \overrightarrow{M_0 A} \quad \overrightarrow{A M_0}$$

$$(4-x_0=1; -3-y_0=2; 1-z_0=-1)$$

$$(3; -5; 2)$$

(11)

~480

δ) $F = (5, 0)$

d: $x = 0$

$F = (\pm \frac{p}{2}, 0)$

$p = 10$

$y^2 = 20x$

c) $M(0, 0)$

$M(1, -4)$

$\begin{cases} p6 = 2p \\ p = 8 \end{cases}$

$y^2 = 16x$

e) $M(6, -2)$

$y^2 = 2px$

$4 = 12p$

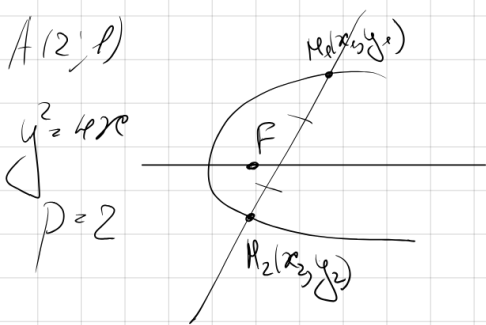
$x^2 = 2py$

$36 = -4p$

$p = -9$

Кривые

~492



$x_1 = \frac{y_1^2}{4}$

$x_2 = \frac{y_2^2}{4}$

$\frac{x_1 + x_2}{2} = 2$

$x_1 + x_2 = 4$

$\frac{y_1 + y_2}{2} = 1$

$y_1^2 + y_2^2 = 16$

$y_1 + y_2 = 2$

$(2 - y_2) + y_2^2 = 16$

$4 - 4y_2 + 2y_2^2 = 16$

$y_2^2 - 2y_2 - 6 = 0$

$D = 4 -$

~496

$y^2 = 12x$

a) $x_0 = 3$

$y^2 = 36$

$y = \pm 6$

$M_0(3, \pm 6)$

?

?

$$\sqrt{375}/b, d) /y/$$

$$q) \text{ напомним } = 4, 2$$

$$a=4$$

$$b=2$$

$$\frac{x^2}{16} + \frac{y^2}{4} = 1$$

$$\delta) |F_1 F_2| = 6, a=5$$

$$F_1(-3, 0) F_2(3, 0)$$

$$c=3$$

$$c^2 = a^2 - b^2 \quad b^2 = a^2 - c^2$$

$$b^2 = 25 - 9 =$$

$$b=4 \quad = 16$$

$$\frac{x^2}{25} + \frac{y^2}{16} = 1$$

$$\sqrt{376}$$

$$\frac{x^2}{169} + \frac{y^2}{25} = 1$$

$$a=13 \quad b=5 \quad \text{очень}$$

$$c^2 = a^2 - b^2$$

$$c = \sqrt{169 - 25} =$$

$$= \sqrt{144} = 12$$

$$F_1(-12, 0)$$

$$F_2(12, 0)$$

$$e = \frac{c}{a} = \frac{12}{13}$$

$$\sqrt{433}/b, d)$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$c=5$$

$$a=4$$

$$c^2 = a^2 + b^2 \quad b=3$$

$$b^2 = 25 - 16 = 9$$

$$\frac{x^2}{16} - \frac{y^2}{9} = 1$$